

Special right triangles



1 The side opposite the angle measuring 30° .

2

a $\frac{\sqrt{3}}{2}$

b $\frac{1}{2}$

c $\frac{1}{\sqrt{3}}$

d $\frac{1}{\sqrt{2}}$

3

a $\sqrt{3}$

b $x = 30^\circ$

c

i $\frac{\sqrt{3}}{2}$

ii $\frac{1}{2}$

iii $\sqrt{3}$

d

i $\frac{1}{2}$

ii $\frac{\sqrt{3}}{2}$

iii $\frac{1}{\sqrt{3}}$

Exact value trigonometric ratios

4

a 60°

b 0.5

5

a 45°

b 1

6

a 30°

b 0.5

7

a 60°

b 1.7

8

a 45°

b 0.71

9

a 30°

b 0.5

10

a 30°

b 0.9

11

a 30°



b 0.58

12

a 60°

b 1.73

13

a 60°

b 0.9

14

a 45°

b 60°

c 60°

d 30°

15

a $a = \sqrt{3}$

b $\theta = 60^\circ$

16

a 45°

b 60°

17

a $x = \frac{\sqrt{30}}{2}$

b $h = 1$

c $x = 2$

d $P = 17$

e $w = 22$

f $x = 4\sqrt{6}, y = 4\sqrt{3}$

g $u = 8, v = 4\sqrt{3}$

h $a = \frac{9\sqrt{3}}{2}, b = \frac{9}{2}$



Applications

18

- a 60°
- b 30°
- c x units
- d $x\sqrt{3}$ units
- e
 - i False. The hypotenuse of a 30° - 60° - 90° triangle is twice the length of the shorter leg.
 - ii True
 - iii False. The longer leg of a 30° - 60° - 90° triangle is $\sqrt{3}$ times the length of the shorter leg.
 - iv True

19

- a
 - i $p = \frac{5\sqrt{3}}{3}$
 - ii $q = \frac{10\sqrt{3}}{3}$
 - iii $r = \frac{5}{2}$
 - iv $s = \frac{5\sqrt{3}}{2}$
- b $\frac{15\sqrt{3}}{2} + \frac{5}{2}$